

MandiMitra

A transparent market-choice assistant for perishable produce

COMMIT FOR GOOD · Round 1 Innovation Brief (draft for team review)

Status: Working prototype with synthetic demo data. Problem validation, real-data testing and team review remain required.

1. Problem discovery

Small-scale producers selling perishable crops must make time-sensitive marketing decisions. A market with a higher quoted/modal price may not yield the highest proceeds after transport expense and potential quality/value loss are considered. Public price records can help, but the decision also depends on a seller's location, harvest size and costs.

Problem hypothesis: Small-scale sellers may struggle to compare nearby markets on expected net proceeds because market price information is separated from seller-specific distance and cost assumptions. This causal link to post-harvest loss must be validated; it is not assumed proven.

2. Problem validation and significance

The Ministry of Food Processing Industries reported, based on NABCONS (2022), estimated post-harvest losses of 4.87–11.61% for vegetables overall; tomato was estimated at 11.61%. These figures establish a broad issue, not evidence that this tool will reduce losses. Source: PIB (2022), <https://www.pib.gov.in/Pressreleaseshare.aspx?PRID=1885038&lang=2®=48>.

WRI India's 2024 Tomato Trail study in Madhya Pradesh identified farm and retail levels as critical loss points and recommended better access to information on price, supply, demand and weather. Source: <https://wri-india.org/research/tomato-trail-tracking-food-loss-and-food-waste-madhya-pradesh>. The Government of India OGD portal publishes daily market price records generated through AGMARKNET: <https://www.data.gov.in/catalog/current-daily-price-various-commodities-various-markets-mandi>.

Still required: interviews with farmers/FPOs/intermediaries; verification of data coverage, access, terms and quality.

3. Existing solutions and gap

AGMARKNET/OGD provides market-price information. e-NAM connects participating APMC mandis and supports electronic trading and price discovery (<https://www.enam.gov.in/web/>). These are important existing services and MandiMitra does not replace them. The proposed gap to test is a lightweight comparison that combines historical prices, market distance, harvest quantity and explicitly adjustable cost/loss assumptions in one view. This is a hypothesis about integration and usability, not a claim that no comparable tool exists. A competitor scan and user interviews are needed.

4. Proposed solution and AI approach

MandiMitra accepts historical market-price CSV data and market distances. For each market, it uses a trailing moving average as a transparent near-term price estimate, calculates estimated gross revenue, subtracts user-entered transport costs and a user-entered linear value-loss scenario, and ranks the results. It compares the recommendation with nearest-market and highest-latest-price heuristics. Forecasts are evaluated on chronological holdout predictions against a last-observation baseline. The moving average is a baseline, not an advanced AI claim; a more complex model is justified only if it improves held-out performance.

5. Innovation and differentiator

The differentiator being tested is a reproducible, explainable decision workflow that converts public market observations into a seller-specific comparison and makes assumptions visible. The team will test whether it adds value beyond price-only or distance-only choices. No claim of novelty is final until the competitor scan is complete.

6. Prototype, evaluation and impact

The current Streamlit prototype supports CSV upload, validation, market ranking, forecast sanity checks and caveats. Bundled data is synthetic and only tests the software. Evaluation plan: obtain a documented real dataset for one crop and bounded market set; compare forecasts against a chronological baseline; compare rankings against nearest-market and highest-price heuristics under varied costs; test an unseen input; seek user feedback. Scenario estimates are not realised income or measured waste reduction. Field trials are needed for real-world impact.

7. AI usage declaration

AI tools assisted with brainstorming, code scaffolding, documentation and test design. The team must identify the tools actually used, review all output and describe suggestions modified or rejected. The team chose a transparent calculation/ranking workflow instead of making a generic LLM chatbot the core product. Update this statement to accurately reflect the actual team process.

References: MoFPI/PIB (2022); NABCONS (2022); WRI India (2024); Government of India OGD/AGMARKNET; e-NAM official website. Full links are in docs/INNOVATION_BRIEF.md and README.md.